

# BioMechAI — Video Cleaning & Deduplication Cycle

How 4,434 Raw Video Clips Were Audited into 3,159 Clean, Zero-Duplicate Clips

AUDITED CLEAN  
CORPUS

## 1. Executive Summary: The Data Problem & Solution

In machine learning for fitness and exercise recognition, raw public datasets are naturally noisy. Web scrapers collect workout videos where repetitions loop, users re-upload identical exercise clips, or athletes stand still between sets. If an AI model is trained on uncleaned data, it memorizes identical videos and stationary background frames rather than true exercise biomechanics.

To build our active **Champion Model PoseC3D v7**, we designed and executed an automated **5-Phase Data Hygiene and Deduplication Pipeline** that audited 4,434 raw clips, removed 987 duplicate clips, and eliminated 284 low-motion noise frames.

## 2. Visual Flowchart: The 5-Phase Cleaning Journey

PHASE 1

**Raw Video Sourcing and Uniform 48-Frame Extraction**  
Collected 4,434 raw clips across 572 distinct video sources. Resampled every clip into uniform 48 frames at stride 2 (2.5 seconds).

PHASE 2

**MD5 Cryptographic Hash Deduplication (Purged 987 Clips)**  
Computed byte-level MD5 hashes over 3D pose coordinate matrices. Eliminated 987 byte-identical duplicate clips.

PHASE 3

**Zero-Leakage Cross-Audit Against Testing Set (Purged 4 Clips)**  
Checked all candidate clips against our held-out test videos. Eliminated 4 overlap clips to guarantee 0.00% data leakage.

PHASE 4

**Kinematic Noise and Stillness Filtering (Purged 284 Clips)**  
Purged 96 static/frozen frames, 157 resting/stationary clips (motion under 5 degrees), and 31 boundary-clipped videos.

PHASE 5

**Final Mathematical Verification (3,159 Clean Training Clips)**  
Enforced mathematical assertions confirming 0.00% duplication and 0.00% leakage across all 7 exercise classes.

## 3. The Exact Mathematics of the Data Cleaning

Raw Candidate Pool: 4,434 clips

- [-] Purged Duplicate Clips: -987 clips (Byte-level MD5 matches)
- [-] Purged Test Overlap: -4 clips (Zero-leakage guard)
- [-] Purged Static / Frozen Frames: -96 clips (Motion variance below 1e-4)
- [-] Purged Resting / Low Motion: -157 clips (Joint Range of Motion below 5 degrees)
- [-] Purged Out-of-Frame Clips: -31 clips (Landmark confidence below 0.35)

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FINAL CERTIFIED CLEAN TRAINING CORPUS: 3,159 CLIPS (100% Unique, Zero Leakage)

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Detailed Phase Breakdown, Accuracy Impact, & Panel Speech Script

PART 2: METHODOLOGY & IMPACT

## 4. Detailed Step-by-Step Phase Explanation

### Phase 1: Uniform 48-Frame Extraction (The 2.5-Second Golden Window)

Different cameras record at 24 FPS, 30 FPS, or 60 FPS. If raw videos are fed directly to an AI, the speed of the video confuses the network. We normalized every video by uniformly sampling **48 frames at stride 2**. This captures exactly 2.5 seconds of physical activity, which corresponds to the biomechanical cadence of one complete exercise repetition.

### Phase 2: Cryptographic MD5 Deduplication

**What is MD5 Deduplication?** In simple words, every video's 3D skeleton coordinates were converted into a digital fingerprint using the MD5 cryptographic algorithm. If two video clips have the exact same numbers, their digital fingerprints are identical:

- We identified that **987 clips were exact copies** of other clips in the dataset.
- We immediately deleted all 987 duplicate copies, keeping only 1 original sample of each movement.
- **Why this was critical:** If duplicates exist, the model simply memorizes the duplicated clips and scores falsely high during training while failing on real human users.

### Phase 3: Train-to-Validation Zero-Leakage Cross-Audit

To ensure our evaluation is academically honest and uncompromised, we cross-checked every single training candidate against our 115 held-out test videos:

- We identified **4 clips** that had visual similarities to test subjects and deleted them.
- **Result:** Guaranteed **0.00% Subject Leakage**. No person, room, or clothing from training ever appears in testing.

### Phase 4: Kinematic Noise and Resting Filter

An exercise recognition model should only learn from active exercise movements. We inspected the joint angles and velocities across all 48 frames:

- **96 Static Frames Purged:** Removed videos where the screen froze, buffered, or showed a gym logo title screen (motion variance below  $1e-4$ ).
- **157 Low-Motion Resting Clips Purged:** Removed clips where an athlete was standing still, drinking water, or resting between sets (joint range of motion under 5 degrees).
- **31 Boundary-Clipped Videos Purged:** Removed videos where the athlete stepped out of frame and their legs or head were cut off.

### Phase 5: The Benchmark Accuracy Impact (53.38% to 63.54%)

Training on this clean corpus transformed our deep neural network:

| EVALUATION DIMENSION | V5 (BEFORE DATASET CLEANING) | V7 (AFTER 5-PHASE CLEAN CORPUS) | ENGINEERING SIGNIFICANCE                           |
|----------------------|------------------------------|---------------------------------|----------------------------------------------------|
| Training Set Size    | 1,720 Raw Clips              | 3,159 Clean Clips               | Expanded clean data by +83.6% with zero duplicates |

| EVALUATION DIMENSION | V5 (BEFORE DATASET CLEANING) | V7 (AFTER 5-PHASE CLEAN CORPUS) | ENGINEERING SIGNIFICANCE                          |
|----------------------|------------------------------|---------------------------------|---------------------------------------------------|
| Duplicate Rate       | 22.2% Duplicates Present     | 0.00% Duplicates (MD5 Verified) | Eliminated dataset memorization                   |
| Top-1 Accuracy       | 53.38%                       | 63.54%                          | + 10.16% absolute accuracy gain on unique clips   |
| Macro Recall         | 53.15%                       | 61.39%                          | Treats all 7 exercises equally without class bias |
| Live App Confidence  | 94.20%                       | 97.58%                          | Decisive real-time inference on mobile phone      |

5. How to Answer the Panel in 20 Seconds (Speech Script)

SPEAK THIS EXACT SCRIPT TO THE EXAMINERS:

"Respected panel, to ensure our AI model learned true human biomechanics rather than memorizing video files, we developed a 5-Phase Data Cleaning Pipeline across 4,434 raw clips. First, we applied MD5 cryptographic hashing to eliminate 987 byte-identical duplicate clips. Second, we cross-audited against our testing set to eliminate 4 overlap clips, guaranteeing 0.00% leakage. Third, we filtered out 284 low-quality samples containing frozen screens, resting athletes, or clipped boundaries. Training PoseC3D on the resulting 3,159 clean clips eliminated dataset memorization, propelling our Top-1 accuracy to 63.54% with a 61.39% Macro Recall on strictly unique test clips."